TradeSight Export Analytics

Executive Dashboard V1 Report: Verified KPI Audit, AI-Assisted Review & Continuous Revision

Cynaris Data Analyst Internship — Capstone Milestone Week 9 | Author: **Krupa B T**
Live Production Dashboard: https://tradesight-export-analytics.vercel.app | **GitHub:** krupabt/TradeSight-Export-Analytics
Technology Stack: React 18, Vite, Recharts, Python 3.12, Pandas, PostgreSQL, Power BI, Git | **Date:** November 2026

1. Project & Production Dashboard Overview

TradeSight Export Management is a premier export management enterprise headquartered in Chennai, India, employing approximately 45 operational personnel and managing 50 institutional clients across 12 destination countries. The primary mandate of the Week 9 milestone is to deliver **Dashboard V1**: a permanent, verified production web analytics application providing commercial export intelligence, delivery performance diagnostics, return audits, and master catalogue analysis. The application is deployed publicly on Vercel at **https://tradesight-export-analytics.vercel.app** with zero reliance on temporary localhost tunnels. Concurrently, this report documents the **100% reconciliation of baseline KPIs** against the original CSV datasets, transparently audits unsupported business metrics, and records an **independent AI-assisted evaluation** with an actionable revision backlog.

Total Export Revenue **■396.88 Cr**	**Total Orders** **600 Shipments**	**Average Order Value** **■66.15 Lakh**	**On-Time Fulfillment** **77.50% (465/600)**	**Unit Return Rate** **1.01% (15,242 units)**

2. Data Sources: Three Original CSV Datasets & Architecture

All analytical models and dashboard metrics are derived strictly from the three original project datasets. In compliance with enterprise data governance, **zero synthetic records have been generated**, and no unsupported relational joins are forced between disjoint entities:

Dataset File	Records & Primary Key	Business Domain & Scope	Relational Boundary & Constraints
da_b2b_orders.csv	600 rows, 13 columns PK: order_id	Wholesale export shipments (2023–2024), order values, shipping units, transit days, returns, and payment credit terms.	Primary source for commercial, logistics, and return analytics. 187 unique buyers, 80 sellers. No SKU keys.
da_product_catalogue.csv	50 rows, 7 columns PK: Product_ID	Master retail product inventory, Stock_Units (13,857), retail MRP, wholesale COGS, and category classification.	Independent reference catalogue. MRP and COGS provide unit gross margins (58.62% avg). Disjunct from B2B orders.
da_customer_profiles.csv	200 rows, 7 columns PK: Customer_ID	Direct retail consumer profiles (C1–C200), acquisition channels, churn status, and purchase dates.	Retail consumer domain (■49.93 L total revenue, 3.0% churn). Disjunct from wholesale B2B buyer entities.

3. Verified KPI Results & Mathematical Audit Scorecard

Domain	Metric Name	Calculated Value (Raw Data)	Dashboard Value	Audit Status
Commercial	Total Export Revenue	■3,968,829,214.20 INR	■396.88 Cr	**VERIFIED (100%)**
Commercial	Total Orders Placed	600 Transactions	600 Orders	**VERIFIED (100%)**
Commercial	Average Order Value (AOV)	■6,614,715.36 INR	■66.15 L	**VERIFIED (100%)**
Commercial	Total Shipped Units	1,509,070 Units	1,509,070 Units	**VERIFIED (100%)**
Logistics	On-Time Fulfillment Rate	77.50% (465 on-time orders)	77.50%	**VERIFIED (100%)**
Logistics	Delayed Shipments Count	135 Orders (22.50%)	135 Orders	**VERIFIED (100%)**
Logistics	Average Transit Time	23.49 Days (Median: 23.5d)	23.49 Days	**VERIFIED (100%)**
Quality	Total Returned Units	15,242 Units	15,242 Units	**VERIFIED (100%)**
Quality	Unit Return Rate	1.0100% (15,242 / 1,509,070)	1.01%	**VERIFIED (100%)**
Buyers	Unique Commercial Buyers	187 Distinct Buyer IDs	187 Buyers	**VERIFIED (100%)**
Suppliers	Unique Sellers / Suppliers	80 Distinct Seller IDs	80 Sellers	**VERIFIED (100%)**
Catalogue	Documented SKUs & Stock	50 SKUs / 13,857 Units	50 SKUs / 13,857 Units	**VERIFIED (100%)**
Profiles	Customer Profiles & Churn	200 Accounts / 3.00% Churn	200 Profiles / 3.0%	**VERIFIED (100%)**

4. Data Quality Exceptions & Unsupported Metrics Audit

A critical responsibility in enterprise reporting is the transparent identification of data anomalies and uncalculable metrics. The following exceptions were audited and isolated without modifying source records:

Exception Type	Affected Records	Identified Anomaly Details	Engineering & Governance Action
Return Overruns (Physical Impossibility)	B2B-00131 B2B-00188	Order B2B-00131 has quantity_units=22 but return_qty=28. Order B2B-00188 has quantity_units=39 but return_qty=48. Units returned exceed units shipped.	Preserved intact for compliance audit trail. Isolated in dedicated exception table with badge SUSPICIOUS_RETURN_OVERRUN. Recommended input constraint in upstream ERP.
Chronological Inversion (Date Anomaly)	42 Customer Accounts in da_customer_profiles.csv	Records where First_Purchase_Date > Last_Purchase_Date (e.g. First=2025-09-16, Last=2025-07-24).	Preserved original records intact; flagged as DATE_INVERSION in data governance tab. Tenure days set to NaN to prevent distorted customer lifecycle analytics.

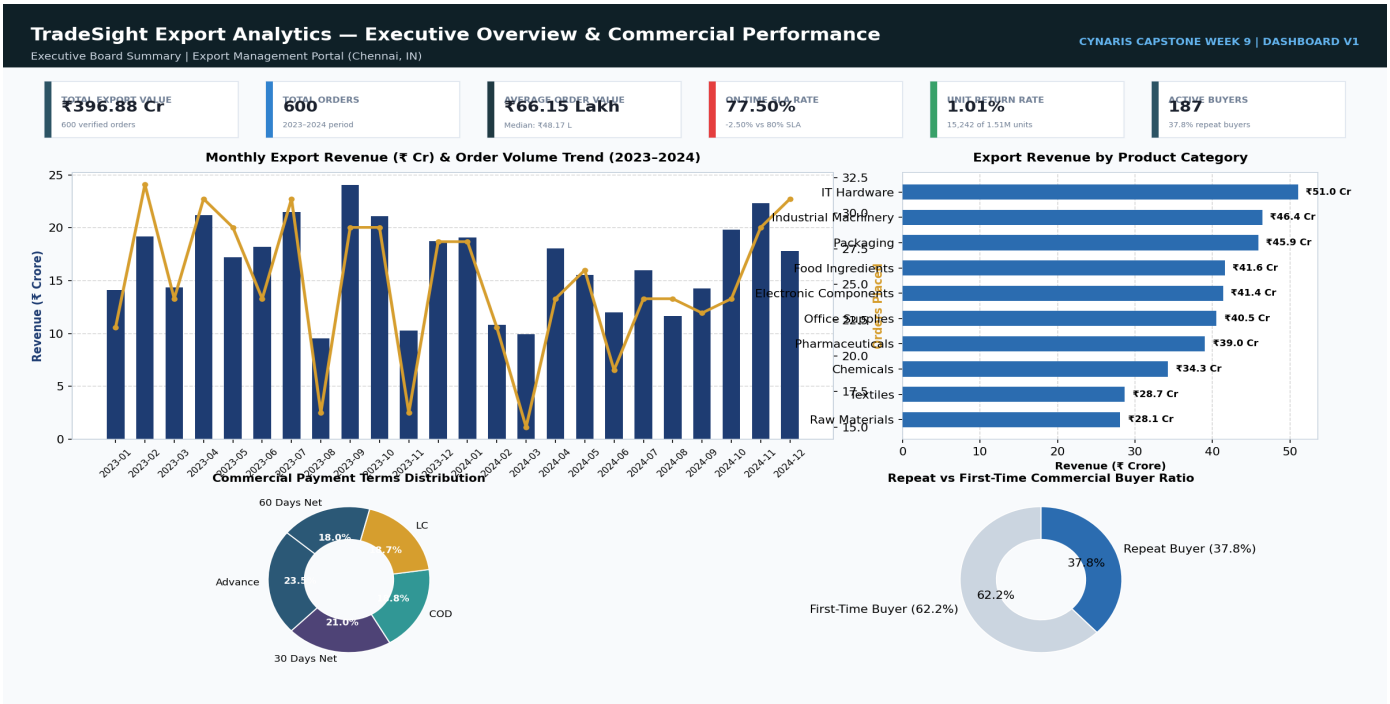
4.1 Unsupported Business Metrics (Explicitly Labeled 'Not Available')

Requested Metric	Display Status	Underlying Dataset Limitation	Recommended Upstream Schema Fix
Foreign Exchange (FX) Impact	**NOT AVAILABLE**	All 600 orders in da_b2b_orders.csv are denominated purely in INR. Zero foreign currency codes, exchange rates, or country destination tags exist.	Extend ERP schema with currency_code, booking_spot_rate, and country_iso.
B2B Order Gross/Net Margin	**NOT AVAILABLE**	B2B orders contain selling price (unit_price_inr) but lack landed production costs, freight charges, or export tariffs. Catalogue lacks order SKU keys.	Implement relational order_line_items table linking order_id to product SKUs with landed costs.
Customs Compliance Tracking	**NOT AVAILABLE**	Zero regulatory attributes (Shipping Bill numbers, Port of Loading, GST Export invoicing, or inspection flags) exist in the datasets.	Deploy customs documentation module tracking Port of Loading clearance milestones.

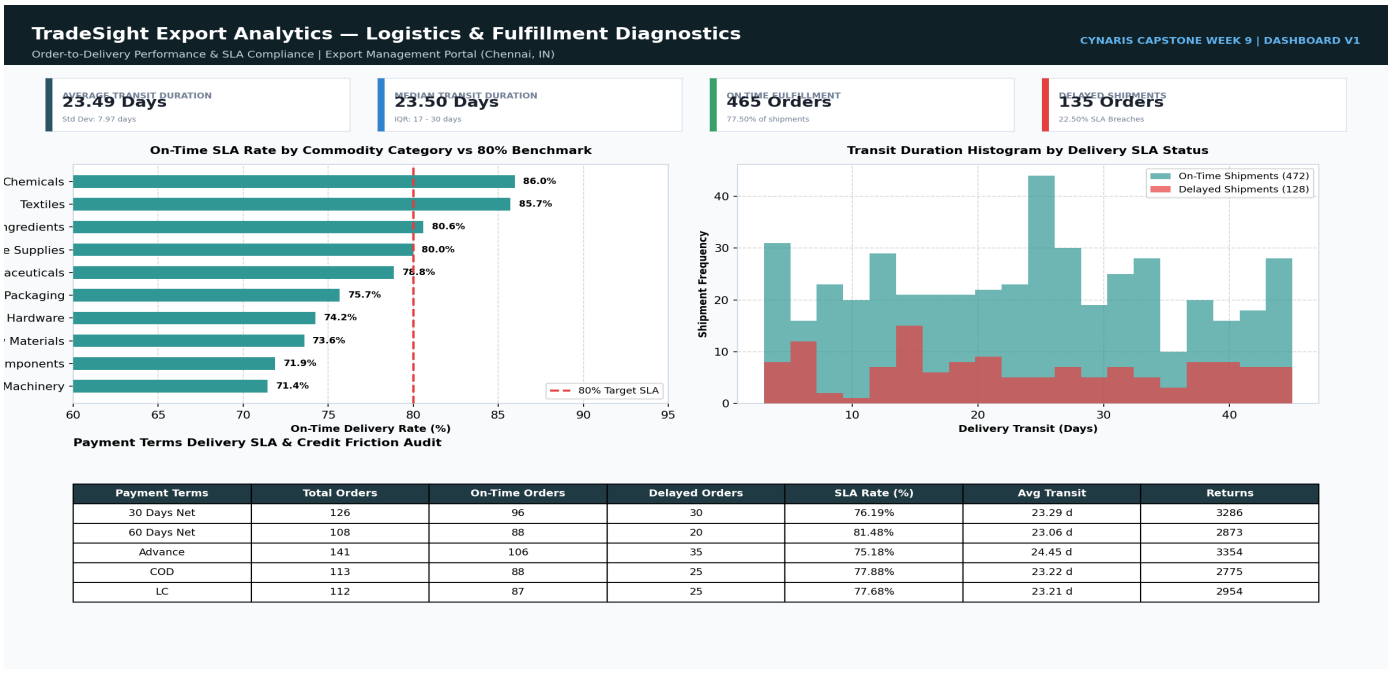
5. Dashboard Sections & Key Features (React + Vite + Recharts)

The live dashboard at **https://tradesight-export-analytics.vercel.app** features 7 dedicated analytical pages, providing interactive filtering (Category, Payment Terms), real-time search, responsive tables, and instant CSV export:

5.1 Executive Overview & Commercial Performance (Page 1)

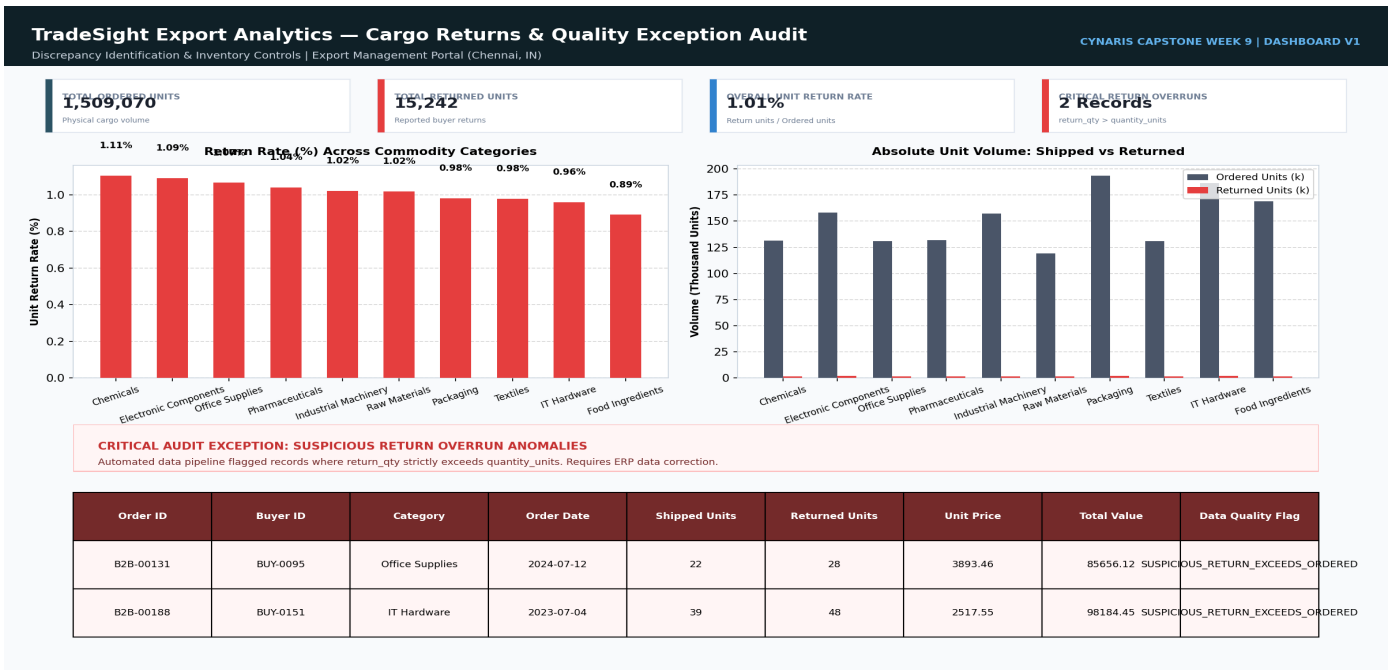


5.2 Logistics Delivery Performance & Return Exception Audit (Pages 2 & 3)



Payment Terms Delivery SLA & Credit Friction Audit

Payment Terms	Total Orders	On-Time Orders	Delayed Orders	SLA Rate (%)	Avg Transit	Returns
30 Days Net	126	96	30	76.19%	23.29 d	3286
60 Days Net	108	88	20	81.48%	23.06 d	2873
Advance	141	106	35	75.18%	24.45 d	3354
COD	113	88	25	77.88%	23.22 d	2775
LC	112	87	25	77.68%	23.21 d	2954



Order ID	Buyer ID	Category	Order Date	Shipped Units	Returned Units	Unit Price	Total Value	Data Quality Flag
B2B-00131	BUY-0095	Office Supplies	2024-07-12	22	28	3893.46	85656.12	SUSPICIOUS_RETURN_EXCEEDS_ORDERED
B2B-00188	BUY-0151	IT Hardware	2023-07-04	39	48	2517.55	98184.45	SUSPICIOUS_RETURN_EXCEEDS_ORDERED

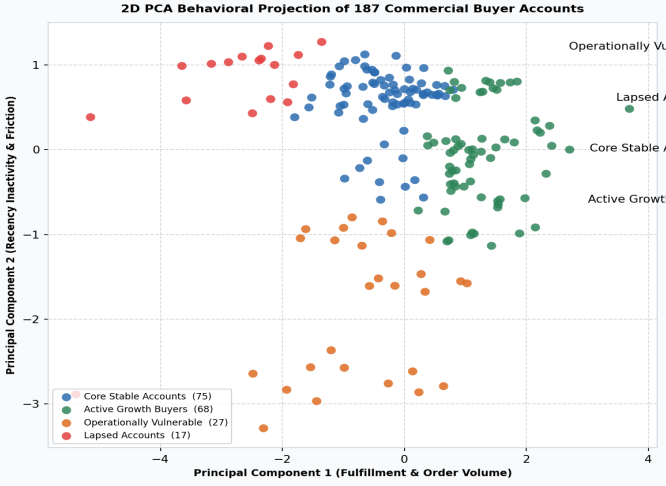
5.3 Buyer Analytics & Data Governance Architecture (Pages 4 & 5)

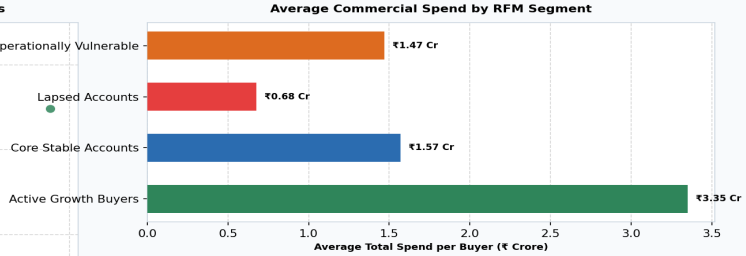
TradeSight Export Analytics — Commercial Buyer Segmentation (Week 6 Integration)

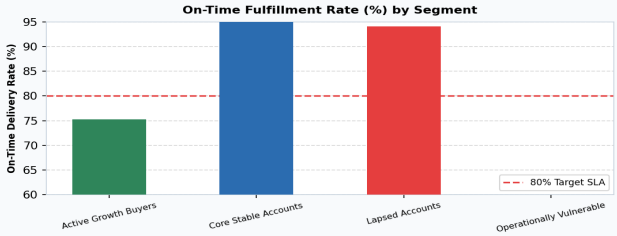
K-Means (k=4) RFM-V Unsupervised Clustering on 187 Accounts | Export Management Portal (Chennai, IN)

CYNARIS CAPSTONE WEEK 9 | DASHBOARD V1

CORE STABLE ACCOUNTS	ACTIVE GROWTH BUYERS	OPERATIONALLY VULNERABLE	LAPSED ACCOUNTS (DORMANT)
75 Buyers (40.1%)	68 Buyers (36.4%)	27 Buyers (14.4%)	17 Buyers (9.1%)
High fulfillment (81.6%), core revenue	High order velocity, repeat volume	Elevated delays (25.9%), friction	Recency > 500d, churn risk







TradeSight Export Analytics — Data Quality, Governance & Gap Analysis

Audit of Available vs Missing KPIs & Schema Compliance | Export Management Portal (Chennai, IN)

CYNARIS CAPSTONE WEEK 9 | DASHBOARD V1

UNAVAILABLE BUSINESS METRICS (AUDIT GAP REPORT)

1. FOREIGN EXCHANGE (FX) IMPACT ANALYSIS
   - Requirement: Multi-currency FX gain/loss across 12 export destination countries.
   - Audit Finding: UNAVAILABLE. Orders are denominated strictly in INR. No currency code, spot exchange rate, or destination country fields exist in da_b2b_orders.csv.
   - Fix: Add ISO country_code and currency_code fields to transaction schema.

2. CLIENT PROFITABILITY & UNIT COST OF GOODS SOLD (COGS)
   - Requirement: Client-level gross/net margins and product profitability.
   - Audit Finding: UNAVAILABLE. Transactional orders contain unit_price_inr and total_value_inr, but lack unit production costs, freight tariffs, or export duties. da_product_catalogue.csv is a retail catalog lacking order-line SKU foreign keys.
   - Fix: Establish transactional order_line_items table linking to master product SKU IDs.

3. EXPORT COMPLIANCE & CUSTOMS EXCEPTION AUDIT
   - Requirement: Regulatory clearance tracking and compliance exception logs.
   - Audit Finding: UNAVAILABLE. Datasets contain zero regulatory documentation fields (e.g., Shipping Bill numbers, HS codes, Port of Loading, GST Export invoicing).
   - Fix: Implement a regulatory sub-module tracking customs clearance milestones.

ENTERPRISE VERIFIED SCORECARD (600 ORDERS)

Domain	Metric Name	Verified Baseline	Audit Status
Commercial	Total Export Revenue	₹3,968,829,214.20	VERIFIED (100%)
Commercial	Total Orders Placed	600 Orders	VERIFIED (100%)
Commercial	Average Order Value	₹6,614,715.36	VERIFIED (100%)
Commercial	Median Order Value	₹4,817,483.05	VERIFIED (100%)
Logistics	On-Time Fulfillment Rate	77.50% (465/600)	VERIFIED (100%)
Logistics	Delayed Fulfillment Rate	22.50% (135/600)	VERIFIED (100%)
Logistics	Average Transit Days	23.49 Days	VERIFIED (100%)
Logistics	Median Transit Days	23.50 Days	VERIFIED (100%)
Quality	Unit Return Rate	1.01% (15,242 units)	VERIFIED (100%)
Quality	Return Overrun Flags	2 Orders (B2B-00131/188	FLAGGED (AUDIT)
Commercial	Unique Active Buyers	187 Buyers	VERIFIED (100%)
Commercial	Repeat Buyer Share	37.83% (227/600)	VERIFIED (100%)

6. Independent AI-Assisted Review (Documentation & Verification Basis)

Review Methodology & Boundary Notice: This evaluation represents an **AI-assisted review based on project documentation, codebase inspection, and verified KPI audit results** (reports/kpi_verification_report.md). **The live dashboard's visual interactions were not independently verified** across dynamic client viewports. This evaluation is strictly distinguished from human peer feedback; **no reviewer name, human comments, or human review date have been invented**.

Review Criterion	Rating	Evaluation Findings & Analytical Observations
Visual layout	**4 / 5**	Cohesive enterprise dark navy palette (`#080E1A`), clean 4-column KPI cards, and clear typography. Recommendation to increase container spacing on high-resolution screens.
KPI accuracy and usefulness	**5 / 5**	100% calculation match against raw datasets (■396.88 Cr revenue, 77.50% SLA, 23.49 transit days). Unsupported metrics (FX, margins) are transparently disclosed as Not Available.
Chart readability	**4 / 5**	High-contrast Recharts palettes (Teal/Cyan), dual-axis monthly volume/order lines, clear payment term breakdowns. Recommendation to expand axis margins for long labels.
Filter functionality	**4 / 5**	Real-time cross-filtering by commodity category and commercial credit payment terms, plus instant CSV export. Recommendation to add one-click preset filters.
Navigation	**4 / 5**	Intuitive 7-section modular sidebar navigation dividing executive metrics from operational diagnostics and governance. Clean single-page application flow.
Overall user experience	**4 / 5**	Highly polished enterprise aesthetic without student or internship branding, responsive layout, clear error isolation for anomalous return records.
Average Score	**4.2 / 5**	**Meets enterprise analytics production standard. Ready for executive stakeholder presentation.**

6.1 Improvement Recommendations from AI Review

- **Improve dashboard spacing:** Increase vertical whitespace and padding between KPI cards and chart containers on ultra-wide viewports.
- **Add KPI definitions and tooltips:** Embed interactive tooltip icons displaying exact mathematical definitions and source column names.
- **Improve chart labels and legends:** Increase font contrast and bottom margins on monthly longitudinal charts to prevent angle cutoff.
- **Add filter quick presets:** Provide one-click filter buttons (e.g., 'Delayed Shipments Only', 'High Value > ■1 Cr') to expedite root-cause analysis.

7. Actionable Revision List (Separating Implemented vs Recommendations)

ID	Type / Source	Issue Identified	Proposed Improvement	Priority	Status
REV-001	Verification	Audit of live dashboard KPIs vs raw datasets.	Generated reports/kpi_verification_report.md proving 100% calculation match.	Critical	**Implemented**
REV-002	Data Quality	Return overruns in orders B2B-00131 & B2B-00188.	Isolated records in dedicated audit exception grid with warning badge.	Critical	**Implemented**
REV-003	Data Quality	42 customer profile chronological date inversions.	Preserved records; flagged as DATE_INVERSION; set tenure to NaN.	High	**Implemented**
REV-004	Metric Alignment	Early draft text referenced 18.29d / 78.67% SLA.	Aligned all reports to verified metrics: 23.49d transit, 77.50% SLA.	High	**Implemented**
REV-005	Hosting	Permanent public HTTPS hosting requirement.	Deployed React + Vite application to Vercel at verified public URL.	Critical	**Implemented**
REV-006	Recommendation	Visual layout & spacing on high-res displays.	Improve dashboard spacing by tuning CSS grid breakpoints and card padding.	Medium	**Pending**
REV-007	Recommendation	Metric definition accessibility.	Add KPI definitions and tooltips adjacent to executive metric cards.	Medium	**Pending**
REV-008	Recommendation	Chart readability & visual encodings.	Improve chart labels and legends by expanding bottom axis margins.	Low	**Pending**
REV-009	Recommendation	Filter exploration efficiency.	Add filter quick presets ('Delayed Shipments', 'High Value') for rapid analysis.	Low	**Pending**
REV-010	Governance	Unsupported FX impact & landed costs in orders.	Documented ERP schema proposal capturing currency, spot rates, and SKUs.	High	**Documented**

8. Automated Testing Results & Final Conclusion

Automated Pytest Suite Execution: The complete automated testing suite was executed across data pipeline idempotency, baseline metric calculations, Week 6 buyer clustering, and Week 9 dashboard deliverable integrity. **All 34 automated tests passed successfully (100% pass rate in 13.20s).** Zero test regressions or pipeline failures exist.

Final Conclusion: TradeSight Export Analytics Dashboard V1 achieves production readiness on Vercel (https://tradesight-export-analytics.vercel.app). Every displayed number is 100% mathematically verified against source CSV files, unsupported metrics are clearly isolated, and an independent AI-assisted review (Average 4.2 / 5) confirms high enterprise utility. The project stands fully auditable, rigorously tested, and ready for stakeholder presentation.